Classificatie Pipeline - Patty Sausele 1146363

```
import numpy as np
import pandas as pd
import sklearn
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_score
import joblib
```

```
df = pd.read_csv('studentset.csv')
df
```

```
                                             transcript  label
0     So when I was a kid ... this was my team.(Laug...      1
1     This is how war starts. One day you're living ...      0
2     In the next 18 minutes, I'm going to take you ...      1
3     Before March, 2011, I was a photographic retou...      0
4     Chris Anderson: And now we go live to Caracas ...      1
...                                                 ...    ...
1845  Most of the time, art and science stare at eac...      0
1846  For emotions, we should not move quickly to th...      0
1847  I published this article in the New York Times...      1
1848  I'll never forget the sound of laughing with m...      1
1849  I'm not at all a cook. So don't fear, this is ...      0

[1850 rows x 2 columns]
```

Data voorbereiden

```
features = df['transcript']
label = df['label']

train_size = int(0.8 * len(df))
X_train, X_test = df['transcript'][:train_size], df['transcript']
[train_size:]
y_train, y_test = df['label'][:train_size], df['label'][train_size:]
```

Bag of Words language model:

```
count_vectorizer = 
CountVectorizer(stop_words='english',min_df=200,max_df=0.95,binary=Tru
e)

X_train_bow = count_vectorizer.fit_transform(X_train)
X_test_bow = count_vectorizer.transform(X_test)
```

```
print(X_train_bow.shape)
print(X_test_bow.shape)
```

```
(1480, 652)
(370, 652)
```

Naive Bayes

```
classificatie_model = MultinomialNB()

classificatie_model.fit(X_train_bow.toarray(), y_train)

predicted = classificatie_model.predict(X_test_bow.toarray())
```

Accuracy Score

```
accuracy_score(y_test, predicted)
```

```
0.5918918918918918
```

Confusion Matrix

```
cm = confusion_matrix(y_test, predicted)
print(cm)
```

```
[[108  75]
 [ 76 111]]
```

Opslaan modellen

```
joblib.dump(classificatie_model, 'classificatie_model.pkl')
```

```
['classificatie_model.pkl']
```

```
joblib.dump(count_vectorizer, 'nlp_model.pkl')
```

```
['nlp_model.pkl']
```